Lec 21 To wrap up time evolution, read S&N 2.2.5.

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Our first elementary solution to a QM problem in 1D (other than free particle).

- Particle in a quadratic potential:

$$\hat{H} = \frac{\hat{p}^2}{2m} + \frac{m\omega^2 x^2}{2}$$

Let's go back to Ehrenfest's theorem:

$$m\frac{d^2\langle x\rangle}{dt^2} = -\langle \nabla V(x)\rangle = -\left\langle \frac{d}{dx}\left(\frac{m\omega^2 x^2}{2}\right)\right\rangle$$

$$\Rightarrow m\frac{d^2\langle x\rangle}{dt^2} = -m\omega^2\langle x\rangle = \text{"F"}$$

force

$$\frac{d^2\langle x\rangle}{dt^2} + \omega^2\langle x\rangle = 0 \Rightarrow \langle x\rangle = A\cos(\omega t) + B\sin(\omega t)$$

oscillatory sol.

Spring analogy

$$\text{"F"} = -m\omega^2\langle x\rangle \quad \text{if } \omega^2 = \frac{k}{m} \Rightarrow \boxed{F = -k\langle x\rangle}$$

Harmonic oscillator

* Bound State
* Discrete Spectrum

Let's learn about the energy of this system and find a solution:

$$E = \langle H \rangle_\psi = \frac{\langle p^2 \rangle}{2m} + \frac{m\omega^2}{2}\langle x^2 \rangle$$

$$\geq \frac{\Delta p^2}{2m} + \frac{m\omega^2}{2}\Delta x^2$$

because $\langle A^2 \rangle = (\Delta A)^2 + \langle A \rangle^2 \geq (\Delta A)^2$

Now let's saturate the uncertainty relation:

$$\Delta p \geq \frac{\hbar}{2}\frac{1}{\Delta x}$$

$$\langle H \rangle \geq \frac{\hbar^2}{8m}\frac{1}{(\Delta x)^2} + \frac{m\omega^2}{2}(\Delta x)^2 \geq \frac{\hbar\omega}{2}$$

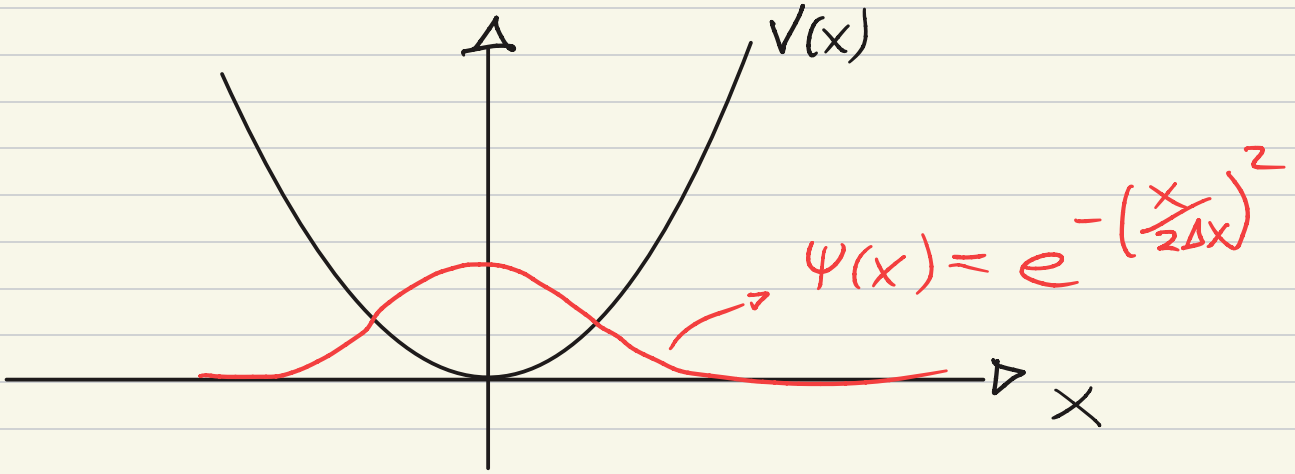
AM-GM inequality: $a + b \geq 2\sqrt{ab}$

Saturate this bound if $\langle x \rangle = 0$; $\langle p \rangle = 0$; and

$$(\Delta x)^2 = \frac{\hbar}{2m\omega}$$

one type of solution is a gaussian

$$\langle x|\psi \rangle = \psi(x) \propto e^{-m\omega x^2/2\hbar} = e^{-\frac{1}{4}\left(\frac{x}{\Delta x}\right)^2}$$



There is a nicer language for this problem. Currently, our $\hat{H}$ mixes canonical pair: $\hat{x}$, $\hat{p}$, both hermitian. It is a sum of squares. For reasons that will soon be evident, we define:

$$a = \sqrt{\frac{m\omega}{2\hbar}}\left(x + \frac{ip}{m\omega}\right) \; ; \quad a^\dagger = \sqrt{\frac{m\omega}{2\hbar}}\left(x - \frac{ip}{m\omega}\right)$$

Calculate: $[\hat{x}, \hat{p}] = i\hbar\mathbb{1}$

$$a^\dagger a = \frac{m\omega}{2\hbar}\left(\hat{x}^2 + \frac{\hat{p}^2}{m^2\omega^2} + \frac{i}{m\omega}\underbrace{(\hat{x}\hat{p} - \hat{p}\hat{x})}\right)$$

$$= \frac{m\omega}{2\hbar}\left(\hat{x}^2 + \frac{\hat{p}^2}{m^2\omega^2} - \frac{\hbar}{m\omega}\mathbb{1}\right) = \frac{\hat{H}}{\omega\hbar} - \frac{1}{2}\mathbb{1}$$

$$\Rightarrow \quad \boxed{\hat{H} = \omega\hbar\left(\hat{a}^\dagger\hat{a} + \frac{1}{2}\right)}$$

Also calculate $[\hat{a}, \hat{a}^\dagger] = \mathbb{1}$.

We call $\hat{N} = \hat{a}^\dagger\hat{a}$ the <u>number</u> operator.

Is it hermitian? Note that $\hat{H} = \omega\hbar\left(\hat{N} + 1/2\right)$

$\hat{N}$ clearly commutes w/ $\hat{H}$ so they can share an eigenbasis. Call it $|n\rangle$:

$$\hat{N}|n\rangle = n|n\rangle \Rightarrow E_n = \left(n + \tfrac{1}{2}\right)\hbar\omega$$

Not clear yet what n or $\hat{N}$ are.
To see the beautiful structure behind them, act $\hat{N}$ on the states $a^\dagger|n\rangle$ and $a|n\rangle$:

$$\hat{N}\,a^\dagger|n\rangle = \left([N, a^\dagger] + a^\dagger N\right)|n\rangle$$

$$= a^\dagger|n\rangle + a^\dagger n|n\rangle$$

$$= (n+1)\,a^\dagger|n\rangle$$

$$\hat{N}\,a|n\rangle = \left([\hat{N}, \hat{a}] + \hat{a}\hat{N}\right)|n\rangle$$

$$= \left(-\hat{a} + \hat{a}n\right)|n\rangle$$

$$= (n-1)\,\hat{a}|n\rangle.$$

The states $\hat{a}^\dagger|n\rangle$ and $\hat{a}|n\rangle$ are still eigenstates of $\hat{N}$ (and also of $\hat{H}$), except eigenvalues have been raised and lowered by one unit.

We call $\hat{a}$ → annihilation operator
$\hat{a}^\dagger$ → creation operator

* Are they hermitian? unitary? Invertible?

Let's first look at the states and fix their normalization.

Given that spectrum is discrete:

$$N|n+1\rangle = (n+1)|n+1\rangle \quad \longleftrightarrow \quad \hat{N}a^\dagger|n\rangle = (n+1)a^\dagger|n\rangle$$

$$N|n-1\rangle = (n-1)|n-1\rangle \quad \longleftrightarrow \quad \hat{N}a|n\rangle = (n-1)|a|n\rangle$$

$$\Rightarrow \quad a|n\rangle = c|n-1\rangle$$

Norm of $a|n\rangle$ is

$$\langle n|a^\dagger a|n\rangle = \langle n|\hat{N}|n\rangle = n$$

$$\text{and} = \langle n-1|c^* c|n-1\rangle = |c|^2$$

$$\Rightarrow \quad \boxed{a|n\rangle = \sqrt{n}\,|n-1\rangle}$$

and

$$a^\dagger|n\rangle = d|n+1\rangle$$

$$\Rightarrow \quad \langle n|aa^\dagger|n\rangle = |d|^2\langle n+1|n+1\rangle = |d|^2$$

$$= \langle n|\hat{N} - [a^\dagger, a]|n\rangle = \langle n|\hat{N} + 1|n\rangle = n+1$$

$$\Rightarrow \quad \boxed{a^\dagger|n\rangle = \sqrt{n+1}\,|n+1\rangle}$$

We recover $\hat{N}$: $\quad a^\dagger(a|n\rangle) = a^\dagger\sqrt{n}\,|n-1\rangle$

$$= \sqrt{n}\sqrt{(n-1)+1}\,|n\rangle$$

$$= n|n\rangle = \hat{N}|n\rangle \checkmark$$

Note that

$$\||n\rangle\| \geq 0 \quad \Rightarrow \quad n \geq 0.$$

$n=0$ is special because

$$a|n=0\rangle = 0 \quad \rightarrow \text{Full Annihilation}$$